SIX

Spacetime Re(con)figurings: Naturalcultural Forces and Changing Topologies of Power

During a transatlantic flight from New York to London, at a cruising altitude of thirty-five thousand feet, a communications link between an Intel-based notebook computer, perched on a tray in front of the passenger in seat 3A of the Boeing 747, and a Sun workstation on the twentieth floor in a Merrill Lynch brokerage house in Sydney initiates the transfer of investment capital from a Swiss bank account to a corporate venture involving a Zhejiang textile mill. The event produces an ambiguity of scale that defies geometrical analysis. Proximity and location become ineffective measures of spatiality. Distance loses its objectivity—its edge—to pressing questions of boundary and connectivity. Geometry gives way to changing topologies as the transfer of a specific pattern of zeros and ones, represented as so many pixels on a screen, induces the flow of capital and a consequent change in the material conditions of the Zhejiang mill and surrounding community. With the click of a mouse, space, time, and matter are mutually reconfigured in this cyborg "trans-action" that transgresses and reworks the boundaries between human and machine, nature and culture, and economic and discursive practices.[1]

The view from somewhere, social location, positionality, standpoint, contextuality, intersectionality, and local knowledges—all are notions that line many a feminist toolbox, for good reasons. And yet these effective and useful tools often implicitly rely on a container model of space and a Euclidean geometric imaginary.[2] The view of space as container or context for matter in motion—spatial coordinates mapped via projections along axes that set up a metric for tracking the locations of the inhabitants of the container, and time divided into evenly spaced increments marking a progression of events—pervades much of Western epistemology. As the geographer Edward Soja points out: "This essentially physical view of space has deeply influenced all forms of spatial analysis, whether philosophical, theoretical or empirical, whether applied to the movement of heavenly bodies or

to the history and landscape of human society. It has also tended to imbue all things spatial with a lingering sense of primordiality and physical composition, an aura of objectivity, inevitability, and reification" (Soja 1989, 79).



Cultural geographers have contested this view of space as a neutral backdrop against which events unfold. A paradigmatic shift occurred with Henri Lefebvre's insistence that space is not a given, but rather that space and society are mutually constituted and that space is an agent of change, that is, it plays an active role in the unfolding of events. Building on David Harvey's theory of geographical historical materialism, Donna Haraway argues that not only class but other material-social practices, such as racialization and gendered sexualization, need to be understood as constituting "bodies-in-the-making and contingent spatiotemporalities" (Haraway 1997, 294). By way of example, Haraway offers the following observation concerning the role that the container model of spatialization plays in the fetishization of gene maps in molecular biology practices:

> Spatialization as a never-ending, power-laced process engaged by a motley array of beings can be fetishized as a series of maps whose grids nontropically locate naturally bounded bodies (land, people, resources—and genes) inside "absolute" dimensions such as space and time. The maps are fetishes in so far as they enable a specific kind of mistake that turns process into nontropic, real, literal things inside containers. (1997, 136)

Haraway's critique of models of spatialization that reify complex practices and make them into things inside containers captures some of the key elements of the kinds of shifts in refiguring space, time, and matter that I am interested in exploring here, including the dynamic and contingent materialization of space, time, and bodies; the incorporation of material-social factors (including gender, race, sexuality, religion, and nationality, as well as class) but also technoscientific and natural factors in processes of materialization (where the constitution of the "natural" and the "social" is part of what is at issue and at stake); the iterative (re)materialization of the relations of production; and the agential possibilities and responsibilities for reconfiguring the material relations of the world. I offer a systematic development and further elaboration of these and related ideas. I consider how agential realism can contribute to a new materialist understanding of power and its effects on the production of bodies, identities, and subjectivities. Central to my analysis is the agential realist understanding of matter as a dynamic and shifting entanglement of relations, rather than a property of things. I develop and explore these ideas in relation to the political theorist Leela Fernandes's ethnographic study of the materialization of the relations of pro-

duction, where questions of political economy and cultural identity formation are both at work on the shop floor.

Following Ruth Wilson Gilmore's suggestion that we replace the politics of location with a politics of possibilities (Gilmore 1999), in this chapter I aim to dislocate the container model of space, the spatialization of time, and the reification of matter by reconceptualizing the notions of space, time, and matter using an alternative framework that shakes loose the foundational character of notions such as location and opens up a space of agency in which the dynamic intra-play of indeterminacy and determinacy reconfigures the possibilities and impossibilities of the world's becoming such that indeterminacies, contingencies, and ambiguities coexist with causality.[3] These reconceptualizations make possible normative analyses crucial to critical political practices without the need for the usual anchor to some conception of fixity.[4] Crucially, these considerations bring into relief the important task of rethinking current conceptions of dynamics (including power dynamics).

Agential realism is an epistemological and ontological framework that cuts across many of the well-worn oppositions that circulate in traditional realism versus constructivism, agency versus structure, idealism versus materialism, and poststructuralism versus Marxism debates.[5] In its reformulation of agency and its analysis of the productive, constraining, and exclusionary nature of naturalcultural practices, including their crucial role in the materialization of *all* bodies, agential realism goes beyond performativity theories that focus exclusively on the human/social realm. Agential realism takes into account the fact that the forces at work in the materialization of bodies are not only social and the bodies produced are not all human. It also provides a way to incorporate material constraints and conditions and the material dimensions of agency into poststructuralist analyses. In these and other important ways, agential realism diverges from feminist postmodern and poststructuralist theories that acknowledge materiality solely as an effect or consequence of discursive practices. These latter approaches lack an account of materiality as an agentive and productive factor in its own right, thereby reinstituting the equation between matter and passivity that some of these approaches proposed to unsettle. Additionally, they leave un(der)theorized a host of pressing questions: What is meant by the claim that discourses have material consequences? What is the relationship between discourse and materiality such that discourse can work its effects? Is there any sense in which materiality might be said to constrain discourses? If so, how? Do material reconfigurations have discursive consequences? What is it about our current material and discursive conditions that questions concerning the material consequences of discourses and the discursive consequences of

materiality seem to preclude each other? This is not to say that such theories do not provide crucial philosophical and political insights. However, critics have noted that they prove inadequate in the face of one of the litmus tests of viable critical social theories: "to *explain* the relation between economic forces—like the formation of new markets through colonization, shifting centers of production, or the development of new technologies—and the reformation of subjectivities" (Hennessy 1993, 25; italics in original). And this is just a short list of material forces that matter.

At the same time, agential realism's reconceptualization of materiality diverges from traditional Marxist conceptions of materiality as strictly economic, and from some post-Marxist conceptualizations that understand it as purely social. Agential realism advances a new materialist understanding of naturalcultural practices that cuts across these well-worn divides.

Leela Fernandes's work also makes significant inroads in this regard, advancing our understanding of social reality in her theorization of the relationship between structural and discursive forces. For Fernandes, as well as for other feminist theorists, like Rosemary Hennessy and Ruth Wilson Gilmore, structural relations are not about structures in the structuralist's sense, and poststructuralism is emphatically not an antidote to Marxism but rather is usefully appropriated as a corrective elaboration of orthodox forms of structural analysis.[6] As these theorists emphasize, understanding class as a dynamic variable with integral cultural, ideological, and discursive dimensions does not diminish, but indeed is necessary to, a thoroughgoing analysis of economic capital in its materiality. Likewise, it is important to recognize the material dimensions of cultural economies.

In this chapter, I diffractively read Fernandes's notion of the structural-discursive relations of power and an agential realist understanding of material-discursive relations of power through each other.[7] This makes it possible to provide a deeper understanding of the nature of structural relations in their materiality and their relationship to discourses, and a new understanding of the dynamics of power relations. It will also suggest a need for remilling some of our most important feminist tools.[8]

PRODUCING WORKERS/PRODUCING STRUCTURES: THE SHOP FLOOR AS A MATERIAL-DISCURSIVE APPARATUS OF BODILY PRODUCTION

Issues of political economy and cultural identity are inseparable. Leela Fernandes's analysis of the structural and ideological workings of power in a Calcutta jute mill gives strong empirical support for this claim. In *Producing*

Workers, Fernandes (1997) employs analytical tools from poststructuralist and Marxist schools of thought, meshing and shifting the gears of these heavy machineries to obtain an understanding of the multiple technologies through which the working class is produced. Disassembling the tenacious assumption in research on labor that "class structure is a uniform, objective 'purity' while other forms of social identity such as gender, religion, and ethnicity are symbolic or ideological forces that either divide or intersect with class identity" (59), Fernandes exposes the manifold connections and detailed intra-(re)workings of identity categories through an examination of shop floor dynamics as they materialize in the course of the everyday lives of the workers.

What motivates Fernandes's study is the following question: Why did the economic crisis in the Indian jute industry in the 1980s result in the differential displacement of women from the jute labor force? Fernandes's topic isn't sexy. It isn't at the forefront of the new technologies. Sewing machines, weaving machines, and other textile machinery line the factory floors; there isn't a DNA-sequencing capillary electrophoresis spectrometer, photolithograph system, or any other device of the new bio-, info-, and nano- technologies anywhere in sight. Fernandes's attention is on what happens on the factory floor, including union politics and the role of religious practices in workers' lives. She hasn't chosen a research site that typifies new capitalist forms of production, distribution, and consumption with their emphasis on service economies, immaterial labor, outsourcing, subcontracting, supply chain economics, flexible accumulation, empire building, deterritorialization, the rhizomic resistance of the multitudes, and the politics of globalization; which is not to say that this is an exceptional site that is walled off from the workings of the latest stage of capitalist production, on the contrary. Fernandes is not interested in building a case for exceptionalism. Indeed, her study does not take exceptionalism (including U.S. exceptionalism) or the dichotomy between the modern and the traditional for granted. Fernandes's study is not a return to the same old venues or the same old methods. Much more is at stake on the shop floor than the production of salable products. The material conditions of the shop floor performatively produce relations of class and other forms of cultural identity in the intra-action of humans and machines. Fernandes's focus is on the detailed dynamics of shop floor relations with an eye toward understanding how production works and doesn't work and for whom. How is difference iteratively produced? What local forms does it take? What differences do differences in production make for the production of different differences?

Fernandes attends to the (re)production of structural relations of differ-

ence by paying close attention to ongoing contests over space, time, and movement in the life of the factory. For example, Fernandes examines the gendered and classed spatialization of the shop floor to look for clues about the ways in which gender and class relations are (re)produced:

> The "structural" dimension of class can be thought of as the ways in which workers are positioned on the factory floor, through recruitment practices and a particular division of labor. This positioning of workers is contingent on the politics of gender and community, since such identities are instrumental in decisions regarding the positioning of workers; thus, gender and community are integral to the class "structure." Meanwhile, the gendering of space signifies particular kinds of class hierarchies between workers and managers and between male and female workers. (Fernandes 1997, 59)[9]

Fernandes uses this examination of the gendering of space to argue that "gender and community are integral to class 'structure' " (59). That is, class itself needs to be understood as "a product of dynamic and contested political processes at the local level of shop-floor politics" (58).

It is important not to mistake this claim for a demotion of class to the realm of the merely ideological/cultural/discursive. A potential misunderstanding of this nature rests on at least three false assumptions that Fernandes calls into question: (1) that economic categories alone are material, and social categories are not; (2) that class is an exceptional identity category; and (3) that identity categories such as class, gender, nationality, caste, and religion are separately determinate attributes of individuals, and an understanding of social dynamics is a matter of knowing how these social factors interact with each other.[10] For Fernandes, class is about economic capital, and at the same time, the "economic" is not merely about class (i.e., the working class is discursively and structurally produced through class, gender, and community). Her shift from a traditional conception of class that assumes that capitalist production is experienced in the same way by all workers all over the world to an understanding of class structures as dynamic and local products goes hand in hand with her insistence that "gender represents a structural force and is not limited to a discursive or symbolic category" (11–12). The gears of the capitalist machinery—which must be understood as different local and contested forms of the global political economy—are simultaneously materially and discursively produced. Fernandes rejects assembly-line notions of identities as analytically identical and interchangeable parts, and she eschews the notion that identities work in lock step as parallel gears in a single assemblage. The dynamics, as

Fernandes describes them, are perhaps more akin to a differential gear assemblage in which the gear operations literally work through one another and yet the uneven distribution of forces results in, and is the enabling condition for, different potentials and performances among the gears.

Fernandes appropriates and extends Foucault's analysis of the important productive effects of disciplinary regimes of power that "partitions as closely as possible time, space, and movement" (Foucault 1977, 137). She argues that "structures" should be understood "as the codification of power through movement, space, and position" (Fernandes 1997, 175), and that "the system of codification that controls time, disciplines movement and partitions space codes workers' bodies through meanings of gender, caste, and ethnicity. If, as Foucault asserts, 'discipline organizes an analytical space' (1979, 143), such techniques of power are in effect employed in the task of producing particular analytical and material borders between class, gender, and community" (Fernandes 1997, 59). I read Fernandes as saying that while disciplinary regimes of power operate through the production of individual subjects, this mode of operation destabilizes, reconfigures, and stabilizes new structural relations of power in reconfiguring the material borders between classes, genders, and communities that mark these very bodies in their materializing subjectivities. In contrast to those who would interpret Foucault's microphysics of power as a refutation of the importance—indeed, the very existence—of structural relations, Fernandes takes Foucault's formulation as an opportunity to rethink the nature and dynamics of structural relations. According to Fernandes, "structure does not represent a set of transcendental, objective determinants but is shaped by modes of representation and meanings that social actors . . . give to their positions and activities" (Fernandes 1997, 137). In other words, structures are not an external set of relations but "force relations immanent in the sphere in which they operate" (Foucault 1978). Furthermore, structures are not only productive; they are themselves produced through the very practices of subject formation that Foucault discusses.

complicating understandings of structuralism

How is Fernandes's claim to be understood? In what sense are structural relations produced and what does this production entail? What is the nature of the processes that "shape" these relations? What is the relationship between the material and discursive dimensions of power relations? How are we to understand the nature of power dynamics? Of materiality? These are some of the questions that I want to explore in diffractively reading Fernandes's powerful insights concerning the structural-discursive relations of power and agential realism through each other.

this is challenging for me to understand too, especially given the status of structuralism as a means for understanding macrosociological processes

I begin by reviewing some key points about agential realism that are particularly relevant to this discussion and then proceed to discuss the example at hand. I propose that the shop floor dynamics be understood in terms of the intra-action of "material-discursive apparatuses of bodily production"—that is, the dynamic intra-workings of the instruments of power through which particular meanings, bodies, and boundaries are produced.[11] Importantly, apparatuses are not external forces that operate on bodies from the outside; rather, apparatuses are material-discursive practices that are inextricable from the bodies that are produced and through which power works its productive effects. Apparatuses are phenomena, material configurations/reconfigurings, that are produced and reworked through a dynamics of iterative intra-activity. This dynamics entails a rethinking of the nature of causality and the role of exclusions in creating the conditions of possibility for contesting and iteratively remaking apparatuses. That is, agential realism does not simply pose a different dynamics (substituting one set of laws for another); it introduces an altogether different understanding of dynamics. What is at issue is not merely that the form of the causal relations are changed, but the very notions of causality, agency, space, time, and matter are all reworked. For example, agency—rather than being thought in opposition to structures as forms of subjective intentionality and the potential for individual action—is about the possibilities for changing the configurations of spacetimematter relations. I discuss the implications of these important shifts in what follows. Using this account of naturalcultural practices makes it possible to attend to the changing "multiplicity of force relations immanent in the sphere in which they operate" (Foucault 1978, 93) where the forces are not merely social, and the bodies produced are not merely human. That is, power is rethought in terms of its overall materializing potential.

breaking down divide of agency and structuralism

TOWARD A POLITICAL ECONOMY OF APPARATUSES, OR HOW APPARATUSES WORK

How do machines work? What kinds of work do machines do? What role do humans play in the operation and production of machines? What role do machines play in the production of other machines and humans and in the reconfiguring of human-machine boundaries and relations? What happens when machines stop working? Could this form of work stoppage be considered a form of machinic agency?

In an article entitled "Mediating Machines," the historian of science Norton Wise contends that machines mediate societal values in the production

of knowledge. He argues, for example, that the steam engine "simultaneously instantiates 'labor value' in political economy and 'work' in engineering mechanics, thereby identifying the two concepts in the region of their common reference." This "partial identification," he claims, "carries with it a structural analogy between a network of concepts from political economy and a similar network in natural philosophy, providing a potent heuristic for the reformulation and further development of dynamics" (Wise 1988, 77). By way of example, Wise begins by pointing to the fact that in 1845, before the development of his work-centered perspective on dynamics (1845–62), William Thomson (Lord Kelvin) began to "regard the idea of natural agency—electric, magnetic, thermal, etc.—as an expression of the capacity to produce work, and thus to regard natural systems as engines" (Wise 1988, 80).

The productive role of apparatuses in linking issues of natural philosophy, political economy, and human and nonhuman forms of agency is one of the central themes of this chapter. However, the analysis that I offer rejects the notion that machines or apparatuses play a "mediating" role. A machine model engineered to explain the influence of social factors on the natural sciences will inevitably be a lopsided device built on a foundational difference between nature and culture. The idea that there are two separate entities or realms of practice influencing one another in determinate regions of overlap is premised on Newtonian conceptions of causality, dynamics, space, and time, and the Newtonian belief in the prior existence of separately determinate bounded and propertied entities and practices. What's missing from this analytical engine is not merely a symmetrical accounting of influence between the natural sciences and political economy, but a model of analysis that isn't a Newtonian instrument.

The shift from Newton's clockwork to Thompson's engine is but a minor mutation when compared to the discontinuous changes that have occurred during the twentieth century in the nature of machines and machinic agency, and our understanding of them.[12] Taking these changes seriously entails a reassessment of the working of all machines, even clockworks, steam engines, and devices that a Newtonian would recognize. It also suggests that the shifts in the epistemological economy of natural philosophy and in the natural philosophy of political economy that have contributed (and continue to contribute) to the production of these changes have more to do with the material entanglement of political and scientific practices than a mediated negotiation between dissimilar systems. I will not offer a historical analysis of these entanglements here. Rather my goal is to put agential realism (an

account of naturalcutural practices which takes seriously insights from some of our best scientific and social and political theories) to work in thinking about the ways in which particular entanglements matter to the production of subjects and objects.

In chapter 3, I presented a detailed exposition of Bohr's analysis of the epistemological work that an apparatus does. Bohr argues that classical physics seriously underestimates and undercounts the contribution that apparatuses make. Apparatuses are not mere instruments serving as a system of lenses that magnify and focus our attention on the object world, rather they are laborers that help constitute and are an integral part of the phenomena being investigated. Furthermore, apparatuses do not simply detect differences that are already in place; rather they contribute to the production and reconfiguring of difference. The failure to take proper account of the role of apparatuses in the production of phenomena seriously compromises the objectivity of the investigation. Accounting for apparatuses means attending to specific practices of differentiating and the marks on bodies they produce.

As I explained in chapter 4, Bohr's account of the apparatus is limited in important respects. Bohr does not attend to important social dimensions of scientific practices, and he fails to offer a consistent account of the role of the subject in these practices and the role of the apparatus in the production of the subject. In that chapter I call upon the insights of social and political theorists to help illuminate particular aspects of the apparatus that Bohr's account leaves unanalyzed. I also discuss some of the limitations of these approaches and consider the possibility of using Bohr's insights to inspire productive emendations and elaborations of these accounts. For example, I argue that while Foucault and Butler attend to the materialization of human bodies as constituted through social forces, they take for granted the materiality of nonhuman beings/bodies and do not consider the productive workings of natural forces. This imbalanced accounting practice translates into an asymmetry in the accounting of material and discursive, natural and cultural, and spatial and temporal, factors in their respective works.[13] These theorists also leave unexamined important ways in which matter is an agentive factor in processes of materialization. My approach is to diffractively read these important insights from natural and social theories through one another in an effort to produce an account of naturalcultural practices and agencies that attends to the production of objects and subjects, the materialization of human and nonhuman bodies, and the entanglement of material relations (including those that get named social, political, economic, natu-

ral, cultural, technological, and scientific, rather than presuming separate factors and domains of operation from the outset).[14] One of the goals, as discussed, is to build an apparatus that is attentive to the nature of specific entanglements. I'm not going to review that analysis here but I do want to highlight some of the significant changes that are entailed in this rethinking of ontology, epistemology, and ethics.

Dynamics are about change. Feminists and other theorists commonly invoke the notion of a power dynamics. In doing so, they often worry about what is meant by power and how it operates, but they assume that the notion of dynamics is a settled and unproblematic concept. Agential realism entails a rethinking of both notions: power and dynamics.

How much of our understanding of the nature of change has been and continues to be caught up in the notion of continuity? For Newton, physicist extraordinaire, inventor of the calculus, author of biblical prophesies, uniter of heaven and earth, continuity was everything, or very nearly. It gave him the calculus. And the calculus gave voice to his vision of a deterministic world: placing knowledge of the future and past at Man's feet. Prediction and retrodiction are Man's for the asking, the price is but a slim investment in what is happening in an instant, any instant. Each bit of matter, whether the size of a planet or an atom, traces out its designated trajectory specified at the beginning of time. Effects follow their causes end on end and each particle takes its preordained place with each tick of the clock. The world unfolds without a hitch. Strict determinism operates like a well-oiled machine. Nature is a clockwork, a windup toy the Omniscient One started up at time t = 0 and then even He lost interest in and abandoned (or perhaps remembers now and again and drops in to do a little tuning up). The universe is a tidy affair indeed. The presumed radical disjuncture between continuity and discontinuity was the gateway to Man's stewardship, giving him full knowability and control over nature. Calculus is the escape hatch through which Man takes flight from his own finitude. Man's reward: a God's eye view of the universe, the universal viewpoint, the escape from perspective, with all the rights and privileges accorded therein. Vision that goes right to the heart of matter, unmediated sight, knowledge without end, without responsibility. Individuals with inherent properties there for the knowing, there for the taking. Matter is discrete but time is continuous. Nature and culture are split by this continuity and objectivity is secured as externality. We know this story well, it's written into our bones, in many ways we inhabit it and it inhabits us.

The quantum disrupts this tidy affair. A bit of a hitch, a tiny disjuncture in the underlying continuum, and causality becomes another matter entirely.

Strict determinism is stopped in its tracks, but the quantum does not leave us with free will either, rather, it reworks the entire set of possibilities made available. Agency and causality are not on-off affairs. This tiny disjuncture, existing in neither space nor time, torques the very nature of the relation between continuity and discontinuity to such a degree that the nature of change changes from a rolling unraveling stasis into a dynamism that operates at an entirely different level of "existence," where "existence" is not simply a manifold of being that evolves in space and time, but an iterative becoming of spacetimemattering (as I explain in chapter 4). Space, time, and matter are intra-actively produced in the ongoing differential articulation of the world. Time is not a succession of evenly spaced intervals available as a referent for all bodies and space is not a collection of preexisting points set out as a container for matter to inhabit. Intra-actions are nonarbitrary nondeterministic causal enactments through which matter-in-the-process-of-becoming is iteratively enfolded into its ongoing differential materialization; such a dynamics is not marked by an exterior parameter called time, nor does it take place in a container called space, but rather iterative intra-actions are the dynamics through which temporality and spatiality are produced and iteratively reconfigured in the materialization of phenomena and the (re)making of material-discursive boundaries and their constitutive exclusions.

The existence of the quantum discontinuity means that the past is never left behind, never finished once and for all, and the future is not what will come to be in an unfolding of the present moment; rather the past and the future are enfolded participants in matter's iterative becoming. Becoming is not an unfolding in time, but the inexhaustible dynamism of the enfolding of mattering.

According to agential realism, causality is neither a matter of strict determinism nor one of free will. Intra-actions always entail particular exclusions, and exclusions foreclose the possibility of determinism, providing the condition of an open future. But neither are anything and everything possible at any given moment. Indeed, intra-actions iteratively reconfigure what is possible and what is impossible—possibilities do not sit still. One way to mark this might be to say that intra-actions are constraining but not determining. But this way of putting it doesn't do justice to the nature of "constraints" or the dynamics of possibility. Possibilities aren't narrowed in their realization; new possibilities open up as others that might have been possible are now excluded: possibilities are reconfigured and reconfiguring.[15] There is a vitality to intra-activity, a liveliness, not in the sense of a new form

of vitalism, but rather in terms of a new sense of aliveness.[16] The world's effervescence, its exuberant creativeness can never be contained or suspended. Agency never ends; it can never "run out." The notion of intra-actions reformulates the traditional notions of causality and agency in an ongoing reconfiguring of both the real and the possible.

In particular, agency is cut loose from its traditional humanist orbit. Agency is not aligned with human intentionality or subjectivity. Nor does it merely entail resignification or other specific kinds of moves within a social geometry of antihumanism. The space of agency is not only substantially larger than that allowed for in most poststructuralist accounts, but also, perhaps rather surprisingly, larger than what liberal humanism proposes. Significantly, matter is an agentive factor in its iterative materialization. Furthermore, the future is radically open at every turn and this open sense of futurity does not depend on the clash or collision of cultural demands. Rather, it is inherent in the nature of intra-activity—even when apparatuses are primarily reinforcing, agency is not foreclosed. Furthermore, the space of agency is not restricted to the possibilities for human action. But neither is it simply the case that agency should be granted to nonhumans as well as humans, or that agency can simply be distributed willy-nilly over nonhuman and human forms.

Crucially, agency is a matter of intra-acting; it is an enactment, not something that someone or something has. Agency is doing/being in its intra-activity. It is the enactment of iterative changes to particular practices—iterative reconfigurings of topological manifolds of spacetimematter relations—through the dynamics of intra-activity. Agency is about changing possibilities of change entailed in reconfiguring material-discursive apparatuses of bodily production, including the boundary articulations and exclusions that are marked by those practices in the enactment of a causal structure. Particular possibilities for (intra-)acting exist at every moment, and these changing possibilities entail an ethical obligation to intra-act responsibly in the world's becoming, to contest and rework what matters and what is excluded from mattering.

As Foucault and Butler emphasize, power is not an external force that acts on a subject; there is only a reiterated acting that is power in its stabilizing and sedimenting effects—only now, in my agential realist account, "the moving substrate of force relations" (Foucault 1978, 92) is not limited to the social.[17] That is, *the forces at work in the materialization of bodies are not only social, and the materialized bodies are not all human*. Furthermore, the productive nature of regulatory and other naturalcultural practices is to be understood in terms

of the causal nature of intra-activity. Crucial to an agential realist conception of power is a reworking of causality as intra-activity. Indeed, what is at issue is the very nature of causal relations: causal relations do not preexist but rather are intra-actively produced. What is a "cause" and what is an "effect" are intra-actively demarcated through the specific production of marks on bodies.

The fundamental discontinuity of quantum physics disrupts the nature of difference: the relationship between continuity and discontinuity is not one of radical exteriority but rather of agential separability, each being threaded through with the other. "Otherness" is an entangled relation of difference. Questions of space, time, and matter are intimately connected, indeed entangled, with questions of justice.

SHIFTING GEARS/SHIFTING DYNAMICS: MANIFOLD POSSIBILITIES FOR THE TOPOLOGICAL RE(CON)FIGURING OF RELATIONS OF POWER

> Class is not this or that part of the machine, but *the way the machine works* . . . the friction of interests—the movement itself, the heat, the thundering noise. . . . class itself is not a thing, it is a happening.
>
> —E. P. THOMPSON, *The Poverty of Theory and Other Essays*

Leela Fernandes's book, *Producing Workers*, is a detailed study of the structural relations of power as they are iteratively (re)produced and contested on the shop floor of a Calcutta jute mill. Fernandes uses the spatial positioning of workers on the shop floor as a material marker of the structural dimensions of class.[18] She cleverly focuses on the material constraints that restrict the positioning and constrain the movement of workers throughout the factory rather than attempting to capture a single deterministic trajectory of power. Indeed, such an idealized trajectory would be meaningless, since it misses the important role that multiple intra-actions, exclusions, and agencies play in the dynamics of power.

In reading Fernandes's work, it is important to notice that material constraints cannot be understood as immutable obstacles in an otherwise unlimited space of freedom. Furthermore, they are not to be interpreted as being completely independent of discursive practices, nor reducible to them, nor as the mere endpoints of such practices. Rather, I read Fernandes's analysis of the dynamics of structural relations in terms of the contingent materialization of the shop floor: not only do the politics of space in the jute mill produce workers as appropriately disciplined subjects in intra-action

with the ever-changing relations of power, but the spatiality of capitalism is itself produced through the politics of gender, community, and class and daily contests over the relations of power by those very subjects. For example, Fernandes argues that "when unions and male workers engage in this reproduction of asymmetrical gender relations, they in fact produce a scattered array of local practices and discourses that maintain the national hegemonic construction of class. In this process, they do not merely use preexisting gendered ideologies but also actively manufacture gender through the creation of particular notions of masculinity and femininity" that wind up reinforcing the powers of management and undermining attempts by the unions to successfully intervene in certain class-based—always already gendered—practices of management (Fernandes 1997, 74). In other words, Fernandes maintains that the spatiality of capitalism is produced not merely through actions of managers who carve up the production process but through the workers' own exclusionary practices as well. That is, while the mill is perhaps most obviously an ongoing process of the materialization of capital, the iterative materialization of the mill is also the outcome of the exclusionary practices of the workers themselves, but not via some linear additive dynamics. Rather, the exclusionary practices of the workers need to be understood to be part of the technologies of capitalism. The intra-action of these material-discursive apparatuses, which includes the practices of the workers as well as the managers, produces a space or structure specifically marked by the topological enfolding of gender, community, and class. In other words, the spatiality of the mill is produced through the dynamics of intra-activity and the reconfiguring and enfolding of structural relations. Structures are apparatuses that contribute to the production of phenomena, but they must also be understood as thoroughly implicated in the dynamics of power: structures are themselves material-discursive phenomena that are produced through the intra-action of specific apparatuses of bodily production marked by exclusions.[19] Structures are specific material configurations/(re)configurings of the world.

Hence, using the framework of agential realism, the jute mill can be understood as an intra-acting multiplicity of material-discursive apparatuses of bodily production that are themselves phenomena materializing through iterative intra-actions among workers, management, machines, and other materials and beings which are enfolded into these apparatuses. Importantly, materiality is rethought as a contingent and contested, constrained but not fully determined, process of iterative intra-activity through which material-discursive practices come to matter, rather than as mere brute positivity or some purified notion of the economic. It is not the case that

economic practices are material, while the presumably separate set of social matters (such as gender and community identity) are merely ideological. The very nature of production is refigured as iterative intra-activity. In the case of the example considered here, this means that production is a process *not merely* of making commodities but also of making subjects, and remaking structures.[20]

Production should not be thought of as the repetition of some fixed set of processes (despite the pervasiveness of the Fordist assembly-line image it often connotes). Rather, the nature of production processes is continually reworked as a result of human, nonhuman, and cyborgian forms of agency. Indeed, as Fernandes points out, when a machine refuses to work, it may initiate a series of events: lost wages for a weaver, a fight between the weaver and the mechanic who was late fixing the machine, the intervention of management to resolve the conflict, union charges against management for mishandling the conflict, a union strike that leads to the restructuring of relations between management and workers, a reconfiguration of machines and workers on the shop floor, or a day off.

Fernandes also observes that the crowding of machines on the shop floor produces the material conditions for workers to crowd together in a way that counters management attempts to institute disciplinary practices that produce individual workers and individualize the nature of work in order to hold individual workers accountable to specified levels of production as established by the management. However, despite management's deployment of a host of low-tech surveillance techniques (e.g., chalkboards and meters installed in the weaving department) that are put in place to track individual worker productivity and hold individuals accountable, the concentration of weaving machines on the shop floor undermines these attempts to individualize the nature of work and provides numerous opportunities for workers to talk with each other. In this way humans and machines together contest the individualization of the nature of production:

> The shop floor tends to be crowded, for machines have been added at various stages in order to increase productivity. . . . The spatial concentration of workers and machines allows workers to talk to each other on the shop floor and has led to numerous complaints by managers that workers tend to gossip and loiter. Such . . . everyday acts of resistance . . . point to the contested nature of the production process and demonstrate that the control of time and movement through the production process represents a political and conflicted terrain. (Fernandes 1997, 63)

Machinic agency is part of the ongoing contestation and reconfiguring of relations of production. The point is not that management and workers become cyborgs in their relationship to machines, but rather the point is that machines and humans differentially emerge and are iteratively reworked through specific entanglements of agencies that trouble the notion that there are determinate distinctions between humans and nonhumans. Workers, machines, managers are entangled phenomena, relational beings, that share more than the air around them; they help constitute one another (e.g., in some cases machines and workers help domestic each other, in other cases they help each other run wild).[21] The entangled, contingent, and changing material conditions of the shop floor produce much more than saleable commodities and the flow of capital is but one stream in a turbulent river of agencies. This shift in theoretical perspective makes visible particular kinds of agency and possibilities for reworking unhealthy and unjust labor conditions that might otherwise be missed if it is assumed that the sole progenitors of agency are human (and only particular humans at that).

Perhaps an elaboration and extension of the differential gear assemblage metaphor that I invoked earlier will provide a useful way of envisioning this understanding of the complex nature of production. The extension that I have in mind is designed to focus attention on the fact that apparatuses are themselves phenomena. Imagine a differential gear assemblage (i.e., a gear assemblage in which the gear operations literally work through one another and in which an uneven distribution of forces results in, and is the enabling condition for, different potentials and performances among the gears) that in an ongoing fashion is being (re)configured/(re)assembled while it is itself in the process of producing other differential gear assemblages. Gears are remilled through intra-actions with other gears, and some gears are in the process of being enfolded into the assemblage as part of its ongoing process of reconfiguration. The assemblages are marked by these processes of (re)assembly. The sedimenting marks of time do not correspond to the history of any individual gear but rather are integrally tied to the genealogy of the assemblage and its changing topology, that is, to the processes of inclusion and exclusion in the reworking of the boundaries of the assemblages.[22] Imagine further that the differential gear assemblages include humans and nonhumans, where the differential constitution of "human" and "nonhuman" changes with each intra-action.[23]

I have engaged this all-too-mechanistic analogy, playing off the most pedestrian metaphor of production, in an attempt to highlight some of the shortcomings of common (mis)conceptions of production processes. For ex-

ample, all too often the focus is either exclusively on the human dimensions of production, distribution, and consumption practices, narrowly conceived as that which occurs in the formal sector of the economy, or on the material culture of these practices in ways that assume separability and stable divisions between the human and the nonhuman.[24] Furthermore, notice that this proposed mutating variant of the machine metaphor for production entails a different understanding of the nature of dynamics—a dynamics in which there is an ongoing reworking of the nature of the production of the very technologies of production themselves. The dynamics of intra-activity are explicitly nonlinear, causal, and nondeterministic. Enfolding is not an arbitrary, random, or automatic process; it is a matter of iterative agential changes in the nature of production. Enfolding changes the topology of spacetime as the connectivity of the spacetime manifold and the boundaries between interior and exterior are reworked. The reconstitution of boundaries and exclusions is an agential process. The apparatuses of production are themselves produced and iteratively reworked, as is the nature of production itself. Agential realism disassembles the notion that structures are Althusserian apparatuses—rigidified social formations of power that foreclose agency and deterministically produce subjects of ideological formations. On the contrary, structures are to be understood as material-discursive phenomena that are iteratively (re)produced and (re)configured through ongoing material-discursive intra-actions.[25]

This machine is not a device assembled out of discrete gears. It would not fit neatly into a Euclidean geometrical framework. It is a topological animal that mutates through an open-ended dynamics of intra-activity. Questions of connectivity, boundary formation, and exclusion (topological concerns) must supplement and inform concerns about positionality and location (too often figured in purely geometrical terms).[26]

As an example, consider the notion of "intersectionality" introduced by feminists of color. Feminists of color who fought hard to displace hegemonic discourses that insisted on the reductive equation "women = gender" tirelessly warned against Euclidean geometrical interpretations of social location and identity formation.[27] Intersectionality, as a well-milled theoretical tool, cuts against the grain of such conceptions. In Fernandes's hands, for example, identity formation is understood not in terms of a Euclidean geometrical model but as a dynamics of changing topologies of space, time, and matter. Identity, in her account, is not about location or positionality with respect to a Euclidean grid of identification. Rather, identity formation is a contingent and contested ongoing material process; "identities" are mutually consti-

tuted and (re)configured through one another in dynamic intra-relationship with the iterative (re)configuring of relations of power. What is so striking about Fernandes's contribution is that her methodological approach enables her to follow the dynamics through which identities and power relations are mutually constituted and iteratively reworked. In particular, she keeps her eye on the way in which the structural relations of production are produced and on the dynamics of the topological reconfiguring of power relations.

Fernandes provides multiple illustrations of the topological nature of these changing dynamics. For example, Fernandes opens her book with an example of the way in which identity categories are produced through one another in following an intra-worker dispute that is transformed into a confrontation between unions and managers and results in a wildcat strike:

> The conflict had begun as a quarrel between two workers on the shop floor. A weaver was waiting for his machine to be fixed by a mechanic. The mechanic did not arrive on time, and the weaver was angry at being unable to work; since his was a piece-rated occupation, the delay had resulted in a loss of wages for the weaver. When the *mistri* (mechanic) finally arrived, an argument started; the mechanic injured the weaver with his hammer, and in the ensuing fight the mechanic was also injured. At this point the general manager and personnel manager . . . took the two to the dispensary. The general manager tried to resolve the conflict, and he made the two workers shake hands. (Fernandes 1997, 1–2)

Fernandes explains that a difference in the caste identities of the weaver and the mechanic played an important role in the union's response to the conflict and the way in which it intra-acted with the factory management:

> The incident reveals the manner in which worker resistance, such as a strike, may arise out of conflicts and social hierarchies between groups of workers. In this case the caste allegiance of the weaver shaped the union's participation and occurred at the expense of the mechanic. However, once the conflict involved a union-management confrontation, it acquired a different meaning for the participants and the workers in general. The wildcat strike rested on a link between the workers' caste positions and union mobilization. However, the meaning of the strike was not limited to this caste relationship. To many workers not involved in the conflict, the strike represented a challenge to an unfair system of authority, that is, within the capitalist system in the factory. In short, there was continual slippage between the politics of caste and class through this sequence of events. (Fernandes 1997, 3–4)

Fernandes notes that in this situation the "unions produced a form of working-class politics that was constructed through caste politics. The boundaries of class interests thus became contingent on caste hierarchies through a specifically political process that involved the participation of workers, unions, and managers in the factory" (Fernandes 1997, 4). But caste hierarchies are themselves produced: "Community identity is created through a conflicted dynamic of hegemony and resistance, a process in which community simultaneously produces and is manufactured through narratives of class and gender within a contested symbolic terrain" (89). Caste, gender, and class materialize through, and are enfolded into, one another. The nature of this enfolding matters to the changing topology, as do iterative changes to the spatiality and temporality of the shop floor, which are constitutive factors in the production of the differential patterns of mattering that constitute the shop floor in its materiality. Structural relations are contingent materialities that are iteratively reworked.[28]

One fascinating thing about Fernandes's analysis is that just when critical social accounts of the workings of power are turning to the postmodern, the posthuman, and the newly emergent, Fernandes returns to the factory floor. She does so not to show the continuing relevance of orthodox Marxist analysis, or simply because this kind of analysis is relevant to what seems to be "old" and not just to what seems to be "new," but because part of what is at stake is the reworking of temporality beyond the usual divisions between the premodern, modern, and postmodern, that is, beyond any developmental sense of temporality.[29] On the other hand, Fernandes's study is also limited in important ways. For starters, Fernandes's genealogies are less attentive to important naturalcultural forces beyond the "bounds" of the factory than what is needed in rethinking questions of scale and the topological reconfigurings that rework the terms of the local and the global.[30] Pushing Fernandes's analysis to the next level would surely entail attending to the workings and production of the causal relations themselves, especially as they are iteratively reconfigured and entangled with other modes of production. I have tried to indicate some of the further complexities of the dynamics that are at issue by considering some theoretical tools that may be useful for this further elaboration, but I have not delved into the specific details of this example and much more work is needed to grasp some of the other relevant entanglements and how they matter and for whom.

What is needed are genealogical analyses not only of the multiple apparatuses of bodily production that come to matter but also of the changing nature of the dynamics itself. Significantly, the agential nature of the iterative

reconfigurings of spacetimematter relations makes clear the need for an ethics of responsibility and accountability not only for what we know, how we know, and what we do but, in part, for what exists.

TOPOLOGICAL MANIFOLDS: SPATIALITY, TEMPORALITY, AND FUTURITY

> How we represent space and time in theory matters, because it affects how we and others interpret and then act with respect to the world.
>
> —DAVID HARVEY, *The Condition of Postmodernity*

The shop floor is not a neutral observing device or a Euclidean frame of reference that allows managers and social scientists to specify the social location of individual workers or to track the trajectories of identity formation. Rather, the apparatuses that make "position" intelligible are implicated in the iterative (re)production of particular material-discursive boundaries among workers. Not only is the notion of position itself a produced, contingent, and contested category that changes through time (not simply whose value changes with time), but "worker" is not a fixed and unitary property of individual human beings, but an actively contested and disunified—but nonetheless objective—category that refers to particular material-discursive phenomena (not individuals). Consequently, it would be inappropriate to view workers as pawns occupying different, but uniform, spaces on the chessboard of an overarching static structure called capitalism; rather, the spatiality of capitalism is itself a contested and ever-changing topology that is iteratively (re)produced through the dynamics of intra-activity and enfolding. The nature of the category "class," its intelligibility and its materiality, depends on these changing dynamics, including intra-actions with particular material-discursive practices that locally define gender and "community." "Thus, 'the working class' does not represent a singular unit but is constituted by status differences" (Fernandes 1997, 10). Likewise, gender, which "represents a type of 'structuring' category, a form of 'habitus' that produces and negotiates patterns within social and cultural life" (11), is itself a contested category whose intelligibility depends in part on the specifics of materializing structural relations (including, for example, ones that might commonly be labeled "economic"). In particular, gender is constituted through class and community and other structural relations of power. Gender, class, and community are enfolded into, and produced through, one another. The claim that class is discursively produced is not a denial of its materiality;

likewise, gender and community are no less material (and no more discursive) than class.

Material conditions matter, not because they "support" or "sustain" or "mediate" particular discourses that are the actual generative factors in the formation of subjects, but because both discourses *and* matter come to matter through processes of materialization and the iterative enfolding of phenomena into apparatuses of bodily production. The material and the discursive are mutually implicated in the dynamics of intra-activity and enfolding. Material and discursive constraints and exclusions are similarly entangled, thereby limiting the validity of analyses that attempt to determine individual effects of material or discursive factors (indeed, they misidentify their objective referents and elide important questions of responsibility). Furthermore, the conceptualization of materiality offered by agential realism makes it possible to take account of material constraints and conditions once again without reinscribing traditional empiricist assumptions concerning the transparency of knowledge practices and the givenness of the world and without falling into the analytical stalemate that simply calls for a recognition of the mediation of the world and then rests its case. The ubiquitous pronouncements proclaiming that experience of the material world is "mediated" have offered precious little guidance about how to proceed. The metaphor of mediation has for too long stood in the way of a more thoroughgoing accounting of the empirical. Incorporating some of the most important insights of poststructuralism, feminist science studies, and other critical reconsiderations of the body, of matter, and of nature, the reconceptualization of materiality offered here makes it possible to take the empirical world seriously once again in the construction and testing of theories, but this time with the understanding that the objective referent is phenomena, not the seeming "immediate givenness" of the world.

In the chapter's opening vignette, I suggest that geometrical analyses are insufficient for a thoroughgoing account of complex events such as the one described. What is the intrinsic metric in this example? What feature unambiguously defines the sense of proximity, location, distance, or scale that determines its geometry? Understanding the dynamics of this complex "trans-action"—which involves not merely the transgression of spatial and other material-discursive boundaries but a re(con)figuration of the spacetimematter manifold itself—requires topological analysis. Questions of size and shape (geometrical concerns) must be supplemented by, and reevaluated in terms of, questions of boundary, connectivity, interiority, and exteriority (topological concerns).

Analyzing the multidimensional, multiply connected, heterogeneous, geopolitical-economic-social-cultural "landscape" on the basis of geometrical considerations will not suffice. Not even if what is meant by geometry is retrofitted for postmodern sensibilities by insisting on the relative and socially constructed nature of presumably geometrical terms (e.g., scale). Nor is it sufficient to figure responsibility in terms of positionality or other efforts to locate oneself within the relevant social horizon. The inadequacy of geometrical analysis in isolation from topological considerations lies in the very nature of "construction." Spatiality is always an exclusionary process, and those exclusions are of agential significance.

For example, in contrast to some unfortunate geometrical readings of the notion of scale (whereby the nesting relationship "local ⊂ national ⊂ global" is presumed to hold in the absence of any critical examination), the geographer Neil Smith explicitly explores the exclusionary nature of the production of scale. He notes that "scale is produced in and through societal activity which in turn produces and is produced by geographical structures of social interaction" (Smith 1992, 62). This insight can be understood in terms of the fact that "scale" refers to a property of spatial phenomena intra-actively produced, contested, and reproduced, and furthermore that it is "an active progenitor of specific social processes" as a result of becoming enfolded into various material-discursive apparatuses of production (66). As Smith emphasizes, "it is precisely the active social connectedness of scales that is vital" (66). This "connectedness" should be understood not as linkages among preexisting nested scales but as the agential enfolding of different scales through one another (so that, for example, the different scales of individual bodies, homes, communities, regions, nations, and the global are not seen as geometrically nested in accordance with some physical notion of size but rather are understood as being intra-actively produced through one another). That is, Smith's notion of "jumping scales" can be elaborated as an element of a topological dynamics in terms of agential enfoldings that reconfigure the connectivity of the spacetimematter manifold.

Boundary transgressions are another instance where geometrical considerations will not suffice. Boundary transgressions should be equated not with the dissolution of traversed boundaries (as some authors have suggested) but with the ongoing reconfiguring of boundaries. For example, information technologies are often touted as the neutrino of the geopolitical-economic-social-cultural landscape, passing through matter as if it were transparent, innocently traversing all borders, whether those of nation-states or different computer platforms, with undiscriminating ease and disregard for obstacles

—the great democratizer, the realization of a mobility and reach that know no bounds. But information technologies do not produce a flat spacetime manifold, a level playing field; on the contrary, in some cases they exacerbate the unevenness of the distribution of material goods, further stabilizing constraints that place restrictions on the everyday lives of those who experience this so-called expansion of opportunity as a diminishing of possibilities.[31] Similarly, as Fernandes (2001) makes clear, trans/nationalism does not make the notion of the nation-state obsolete. The relationship between the local, the regional, the national, and the global is not a geometrical nesting. Local, regional, national, and global are topological matters, intra-actively produced through one another, so that an increase in the flow of information and goods across national boundaries does not in and of itself constitute the obsolescence of the nation-state.

What is needed are genealogies of the material-discursive apparatuses of production that take account of the intra-active topological dynamics that iteratively reconfigure the spacetimematter manifold. In particular, it is important that they include an analysis of the connectivity of phenomena at different scales.[32] As Ruth Wilson Gilmore points out, it is crucial to trace the "frictions of distance," to do analyses that move through the range of scales of injustice, not by pointing out similarities between one place or event and another, but by understanding how those places or events are made through one another.[33]

The topological dynamics of space, time, and matter are an agential matter and as such require an ethics of knowing and being: intra-actions have the potential to do more than participate in the constitution of the geometries of power; they open up possibilities for changes in its topology and dynamics, and as such, interventions in the manifold possibilities made available reconfigure both what will be and what will be possible. The space of possibilities does not represent a fixed event horizon within which the social location of knowers can be mapped, nor does it represent a homogeneous, fixed, uniform container of choices. Rather, the dynamics of the spacetime manifold are iteratively reworked through the inexhaustible liveliness of the manifold's material configuration, that is, the ongoing dance of agency immanent in its material configuration. The politics of identity and the politics of location, however useful, have been circumscribed by a geometrical conception of power that arrests and flattens important features of its dynamics. Perhaps what is needed is a politics of possibilities (Gilmore): ways of responsibly imagining and intervening in the configurations of power, that is, intra-actively reconfiguring spacetimematter.